



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
End Semester Examination
Academic Year: 2025-26 (Semester I)

Class: TY	Subject: Manufacturing Processes - II	Subject Code: PCC-ME304
Date: 18- Nov-2025	Time: 10:00 to 12:00	Max. Marks: 60

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Solve all the questions	Marks	CO	BT
Q.1	Select the manufacturing processes for the given material and justify your answer with a scientific reason a) Frozen Fish Cutting b) Glass Cutting	10	CO6	BT4
Q.2	Draw the schematic of drill bit explain all terminologies related to that. List down the types of drilling machine in detail.	10	CO2	BT1
Q.3	Elaborate Any two Super Finishing processes in detail with diagram, advantages, disadvantages and application	10	CO5	BT2
Q.4	During turning a steel rod of diameter 150 mm by a carbide tool of geometry; 0° , -12° , 8° , 6° , 15° , 60° , 0 (mm) at speed 560 rpm, feed 0.32 mm/rev. and depth of cut 4.0 mm the followings were observed $P_Z = 1000$ N, $P_Y = 200$ N, $a_2 = 0.8$ mm. Determine, without using MCD, the expected values of F , N , μ , P_s , P_n , τ_s , cutting power and specific energy requirement for the above mentioned machining operation.	10	CO1	BT3
Q.5	Determine T_c for plain milling a rectangular surface of length 100 mm and width 50 mm by a helical fluted plain HSS milling cutter of diameter 60 mm, length 75 mm and 6 teeth. Assume $A = O = 5$ mm, $V_c = 40$ m/min and $s_o = 0.1$ mm/tooth	10	CO3	BT3
Q.6	Composition of a Nickel superalloy is as follows: Ni = 70.0%, Cr = 20.0%, Fe = 5.0% and rest Titanium Calculate rate of dissolution if the area of the tool is 1500 mm ² and a current of 2000 A is being passed through the cell. Assume dissolution to take place at lowest valency of the elements.	10	CO4	BT3

$A_{Ni} = 58.71$	$\rho_{Ni} = 8.90$	$v_{Ni} = 2$			
$A_{Cr} = 51.99$	$\rho_{Cr} = 7.19$	$v_{Cr} = 2$			
$A_{Fe} = 55.85$	$\rho_{Fe} = 7.86$	$v_{Fe} = 2$			
$A_{Ti} = 47.90$	$\rho_{Ti} = 4.51$	$v_{Ti} = 3$			

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Demonstrate the metal cutting process and select the appropriate cutting tools for it
CO 2	Exhibit the characteristic, terminologies related to cutting tools.
CO 3	Calculate the machining time, depth of cut, feed required.
CO 4	Demonstrate knowledge related to various machining processes.
CO 5	Exhibits the knowledge of various finishing processes.
CO 6	Select proper machining processes for the product to be manufacture



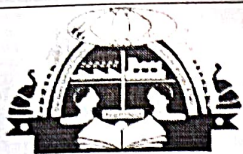
Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
End Semester Examination
Academic Year: 2025-26 (Semester - I)

Class: TV B.Tech.	Subject: Design of Machine Elements	Subject Code: PCC-ME-303
Date: 24.11.2025	Time: 10:00 to 12:00	Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
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- Figures to the right indicate full marks and solve accordingly.
- The symbols carry the usual meaning.
- Mention the question numbers properly while answering.

Q.N.		Marks	CO	BT
Q.1	Solve A and B OR only C.			
	A Draw S-N curve in terms of Low cycle and High cycle Fatigue.	5	CO1	BT1
	B Explain the concept 'Design Consideration for Dynamic and Static Load.'	5	CO1	BT2
	OR			
	C Design a knuckle joint for a tie rod of a circular section to sustain a maximum pull of 70KN. Design a knuckle joint for a tie rod of a circular section to sustain a maximum pull of 70 KN. The ultimate strength of the material of the rod against tearing is 420 MPa. The ultimate tensile and shearing strength of the pin material are 510 MPa and 396 MPa respectively. Determine the tie rod section and pin section. Take factor of safety = 6.	10	CO1	BT6
Q.2	Solve A and B OR A and C.			
	A Draw Soderberg, Goodman line and Gerber parabola in terms of fluctuating stress.	5	CO2	BT3
	B A bar of a circular cross section is subjected to alternating tensile forces varying from a minimum of 200KN to a maximum of 500KN. It is to be manufactured of a material with an ultimate tensile strength of 900Mpa and an endurance limit of a 700Mpa. Determine the diameter of a bar, using safety factors of 3.5 related to ultimate tensile strength and 4 related to endurance limit and a stress concentration factor of 1.65 for fatigue load. Use Goodman straight line as a base for design.	5	CO2	BT6
	OR			
	C Determine the diameter of a circular rod made of ductile material with fatigue strength 400 MPa and tensile yield strength of 550 MPa. The member is subjected to a varying axial load from $W_{min} = -500 \times 10^3 N$ to $W_{max} = 900 \times 10^3 N$ and stress concentration factor is 1.8 with factor of safety 2.	5	CO2	BT6
Q.3	Solve A and B OR A and C.			
	A What is coupling? Classify them and write their advantages and disadvantages. (1+2+1+1=5 Marks)	5	CO3	BT2
	B A solid steel shaft of transmitting power of 20KW at 500rpm. If the maximum torque exceeds mean torque by 30%, calculate the diameter of shaft. The shaft is subjected to a twist of 1°(degree) in a length of 2m. Take $G = 0.84 \times 10^5 N/mm^2$.	5	CO3	BT3
	OR			
	C Design a rectangular key for a shaft 40mm diameter. The shearing and crushing stresses for the key material are 30Mpa and 60Mpa.	5	CO3	BT6
Q.4	Solve A OR B			
	For supporting a crane in a workshop, the brackets are fixed on steel column as shown in figure. The maximum load is 12KN acting vertically downward. The vertical face of the bracket is secured to the column by means of 4 bolts (2 bolts on each row). Determine the size of the bolts if permissible tensile stress is 84Mpa. Also find cross-section of the arm of bracket which is rectangular.	10	CO4	BT6
Q.N.		Marks	CO	BT

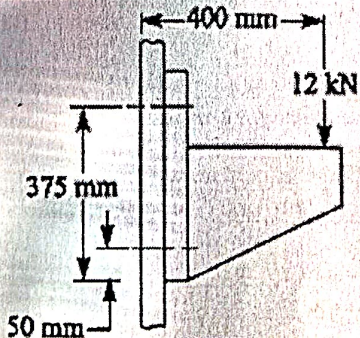


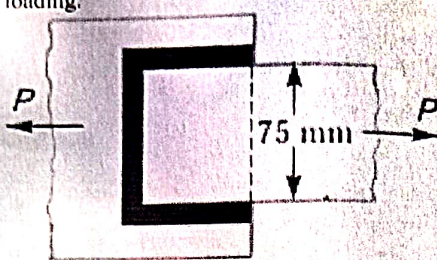
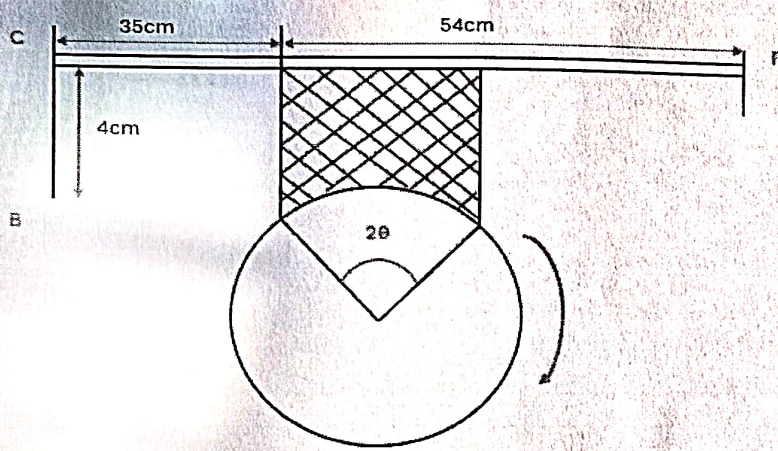
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	OR			
	C Determine the diameter of a circular rod made of ductile material with fatigue strength 400 MPa and tensile yield strength of 550 MPa. The member is subjected to a varying axial load from $W_{min} = -500 \times 10^3 N$ to $W_{max} = 900 \times 10^3 N$ and stress concentration factor is 1.8 with factor of safety 2.	5	CO2	BT6
Q.3	Solve A and B OR A and C.			
	A What is coupling? Classify them and write their advantages and disadvantages. (1+2+1+1=5 Marks)	5	CO3	BT2
	B A solid steel shaft of transmitting power of 20KW at 500rpm. If the maximum torque exceeds mean torque by 30%, calculate the diameter of shaft. The shaft is subjected to a twist of 1° (degree) in a length of 2m. Take $G = 0.84 \times 10^5 N/mm^2$.	5	CO3	BT3
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Q.N.		Marks	CO	BT

Q.4	B	OR	10	CO4	BT6
		A plate 75mm wide and 12.5mm thick is joined with another plate by a single transverse weld and double parallel fillet weld as shown in figure. The maximum tensile stress and shear stresses are 70MPa and 56MPa respectively. Find the length of each parallel fillet weld, if the joint is subjected to both static and fatigue loading. 			
Q.5		Solve A and B OR only C.			
	A	Explain Centrifugal Clutch with its advantages and disadvantages.	5	CO5	BT2
	B	A single plate clutch with its both side effective, has outer and inner diameter 300mm and 200mm respectively. The maximum intensity of a pressure at any point in the contact surface is not to exceed 0.1N/mm ² . If the coefficient of friction is 0.3, determine the power transmitted by a clutch at a speed 2500rpm.	5	CO5	BT6
		OR			
	C	Derive an Equation for Torque transmitting Capacity of Cone clutch using 1. Uniform Pressure and 2. Uniform Wear Theory	10	CO5	BT4
Q.6		Solve A and B OR only C.			
	A	Explain Air brake system with neat sketch.	5	CO6	BT2
	B	Draw and explain Rope Brake dynamometer.	5	CO6	BT3
		OR			
	C	A Simple block brake as shown in figure with a diameter of brake drum is 30cm and angle of contact is 45°. An effort of 600 N is applied at the end of lever. Determine the torque transmitted, if the fulcrum is at B for the clockwise rotation of drum. Take $\mu = 0.4$. 	10	CO6	BT6

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Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Students understand the need, steps of machine design, and consideration for manufacturing
CO 2	Student will be able to evaluate life of machine components against static loads and fluctuating loads.
CO 3	Student demonstrates skill of designing various temporary and permanent joints.
CO 4	Students will design the transmission shafts, key and coupling
CO 5	Students will design the frictional elements like Clutches
CO 6	Students will design the Brakes and dynamo meters.